# EE 5239 Nonlinear Optimization

# Lecture 14a: The Alternating Direction Method of Multipliers (I)

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# Outline

# Last Time

- Last few lectures introduced ways to impose structure in the solution
- Extremely useful when problem size becomes large – low-dimensional representation of high-dimensional data
- Adding penalty to the loss function:
  1. Sparsity –$\|\cdot\|_0$ norm
  2. Group sparsity – $\|\{\|\boldsymbol{x}_g\|_2\}_{g\in G}\|_0$ norm
  3. Low rank– $\text{Rank}(\mathbf{X})$
- These norms are difficult to deal with (even discontinuous)
- Use their convex surrogates
  1. Sparsity –$\|\cdot\|_1$ norm
  2. Group sparsity – $\|\{\|\boldsymbol{x}_g\|_2\}_{g\in G}\|_1 = \sum_{g\in G}\|\boldsymbol{x}_g\|_2$ norm
  3. Low rank– $\|\mathbf{X}\|_*$

# Last Time

- These surrogates are nonsmooth but convex functions
- They share an important property: computing the so-called proximity operator is easy

$$\text{prox}_R^\gamma(\mathbf{y}) = \arg\min_{\boldsymbol{x}} \frac{1}{2}\|\boldsymbol{x} - \mathbf{y}\|^2 + \gamma R(\boldsymbol{x})$$

- Then we can use first-order method to solve the resulting problems
  1. Perform a simple gradient step
  2. Compute the proximity operator on the resulting objects

see `https://www.stat.cmu.edu/~ryantibs/convexopt/lectures/prox-grad.pdf`

# The Anomaly Detection Problem?

$$\begin{aligned} \min \quad & \|\mathbf{L}\|_* + \gamma_1 \|\mathbf{S}\|_1 + \gamma_2 \|\mathbf{Z}\|_F^2 \\ \text{subject to} \quad & \mathbf{L} + \mathbf{S} + \mathbf{Z} = \mathbf{M} \end{aligned}$$

$$\min \quad \|\mathbf{L}\|_* + \gamma_1 \|\mathbf{S}\|_1 + \gamma_2 \|\mathbf{M} - \mathbf{L} - \mathbf{S}\|_F^2$$

1. How to solve it?

# This Time

- Introduce a family of algorithm called Alternating Direction Method of Multipliers (ADMM)
- Popular algorithm for large-scale machine learning, distributed computation, etc.
- Lots of applications
- Our discussion will be based on the following review paper: "Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers", S. Boyd, N. Parikh, E. Chu, B. Peleato, J Eckstein, *Foundation and Trends in Machine Learning*, 2011

# Outline

# Motivation

- **An Ancient Strategy**
- "Divide et impera" – Julius Caesar (100- 44 BC)
- "Divide and Rule" – Sun Tzu, ART OF WAR (535-470 BC)
- In mathematical terms: Split and Alternate

# The Problem to be Solved

- The ADMM algorithm solves the following convex program

$$\begin{aligned} \min \quad & f(\boldsymbol{x}) + g(\mathbf{z}) \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} + \mathbf{B}\mathbf{z} = \mathbf{c} \\ & \boldsymbol{x} \in X, \quad \mathbf{z} \in Z \end{aligned}$$

- $\boldsymbol{x} \in \mathbb{R}^n$ and $\mathbf{z} \in \mathbb{R}^m$ are the variables
- $f$, $g$ are two convex functions, possibly nonsmooth
- $\mathbf{A}$, $\mathbf{B}$ are two known matrices, $\mathbf{c}$ is a known vector
- **Note**: Two blocks of variables; separable in the objective, coupled by a linear equation
- **Main Benefit**: Capable of dealing with $\boldsymbol{x}$ and $\mathbf{z}$ separately (in a BCD manner)

# Application 1: Constrained $\ell_1$ problem

$$\begin{aligned} \min \quad & \|\boldsymbol{x}\|_1 \\ \text{s.t.} \quad & \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|^2 \leq \delta \end{aligned}$$

- Similar as the LASSO, but with explicit constrains on the noise magnitude
- Formulate into the ADMM form?
- Introduce a new variable $\mathbf{z}$ (**split**!), arrive at an equivalent formulation

$$\begin{aligned} \min \quad & \|\boldsymbol{x}\|_1 \\ \text{s.t.} \quad & \|\mathbf{z}\|^2 \leq \delta \\ & \mathbf{z} = \mathbf{A}\boldsymbol{x} - \mathbf{b} \end{aligned}$$

- Mapping back to the original problem, $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_1$, $g(\mathbf{z}) = 0$
- $\mathbf{B} = -\mathbf{I}$, $Z := \{\mathbf{z} \mid \|\mathbf{z}\|^2 \leq \delta\}$, $X = \mathbb{R}^n$

# Application 2: Multiple Regularizations

- Consider the following sparse group LASSO problem
- The problem is formulated as

$$\min \frac{1}{2}\|\mathbf{A}\boldsymbol{x}-\mathbf{b}\|^2 + \underbrace{\sum_{g=1}^{G}\lambda_g\|\boldsymbol{x}_g\|_2}_{\ell_1/\ell_2\text{norm}} + \underbrace{\lambda\|\boldsymbol{x}\|_1}_{\ell_1\text{norm}}$$

- Select a few groups, and select a few variables within groups
- How to deal with two different nonsmooth penalizations simultaneously?

# Application 2: Multiple Regularizations

- Same trick, **split and alternate**
- Introduce two new variables $\mathbf{z}_1$, $\mathbf{z}_2$
- Perform the following reformulation

$$\min \quad \underbrace{\sum_{g=1}^{G} \lambda_g \|\boldsymbol{x}_g\|_2}_{f(\boldsymbol{x})} + \underbrace{\frac{1}{2}\|\mathbf{A}\mathbf{z}_2 - \mathbf{b}\|^2 + \lambda\|\mathbf{z}_1\|_1}_{\mathrm{g}(\mathbf{z})}$$

$$\text{subject to} \quad \mathbf{z}_1 = \boldsymbol{x}, \quad \mathbf{z}_2 = \boldsymbol{x}$$

- Reformulated in an ADMM form; deal with two norms separately?

# Application 3: Intersection of Two Sets

- Suppose we would like to solve the following problem

$$\begin{aligned} & FIND\ \boldsymbol{x} \\ \text{subject to} \quad & \|\boldsymbol{x}\|_1 \leq \delta, \quad \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|^2 \leq \epsilon \end{aligned} \tag{0.1}$$

- Finding the intersection of two convex sets
- We can introduce two variables $\mathbf{z}_1$, $\mathbf{z}_2$

$$\begin{aligned} & FIND\ \boldsymbol{x} \\ \text{subject to} \quad & \|\mathbf{z}_1\|_1 \leq \delta, \quad \|\mathbf{z}_2\|^2 \leq \epsilon \\ & \mathbf{z}_1 = \boldsymbol{x}, \quad \mathbf{z}_2 = \mathbf{A}\boldsymbol{x} - \mathbf{b} \end{aligned}$$

- First block $\boldsymbol{x}$, second block $(\mathbf{z}_1, \mathbf{z}_2)$; Fix $\boldsymbol{x}$, $\{\mathbf{z}_i\}$'s are independent!
- Dealing with each constraint separately

# Application 4: Consensus Problem

- Consider the simple least square problem

$$\min \quad \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|^2 \tag{0.2}$$

- Suppose $\mathbf{A}$ is divided into $N$ groups of rows $\mathbf{A}_1, \cdots, \mathbf{A}_N$
- There are $N$ computing nodes, each obtaining a single piece of data $\mathbf{A}_n$
- The problem can be re-written as

$$\min \quad \sum_{i=1}^{N} \|\mathbf{A}_i\boldsymbol{x} - \mathbf{b}_i\|^2 \tag{0.3}$$

# Application 4: Consensus Problem

- Distributed computation?
- Let us introduce $N$ variables $\mathbf{z}_1, \cdots, \mathbf{z}_n$
- Consider the following reformulation

$$\begin{aligned} \min \quad & \sum_{i=1}^{N} \|\mathbf{A}_i \mathbf{z}_i - \mathbf{b}_i\|^2 \\ \text{subject to} \quad & \mathbf{z}_i = \boldsymbol{x}, \ \forall \ i \end{aligned}$$

- Two-blocks: $\boldsymbol{x}$, $(\mathbf{z}_1, \cdots, \mathbf{z}_n)$
- Each $\mathbf{z}_i$ can be handled by the $i$-th local agent (together with $\mathbf{A}_i$)!
- Fix $\boldsymbol{x}$, $\{\mathbf{z}_i\}$'s are independent!

# Application 4: Consensus Problem

- More generally, consider the following problem

$$\min \quad \sum_i f_i(\boldsymbol{x}) + R(\boldsymbol{x}), \text{ subject to } x \in X \tag{0.4}$$

- We can perform the same procedure as before

$$\begin{aligned} \min \quad & \sum_i f_i(\mathbf{z}_i) + R(\boldsymbol{x}) \\ & \text{subject to } \boldsymbol{x} \in X,\ \mathbf{z}_i = \boldsymbol{x},\ \forall\ i \end{aligned}$$

- Each agent handles a single function $f_i$
- Required to reach a global consensus
- Not necessarily to have the "star network"

# Consensus SVM (from Boyd)

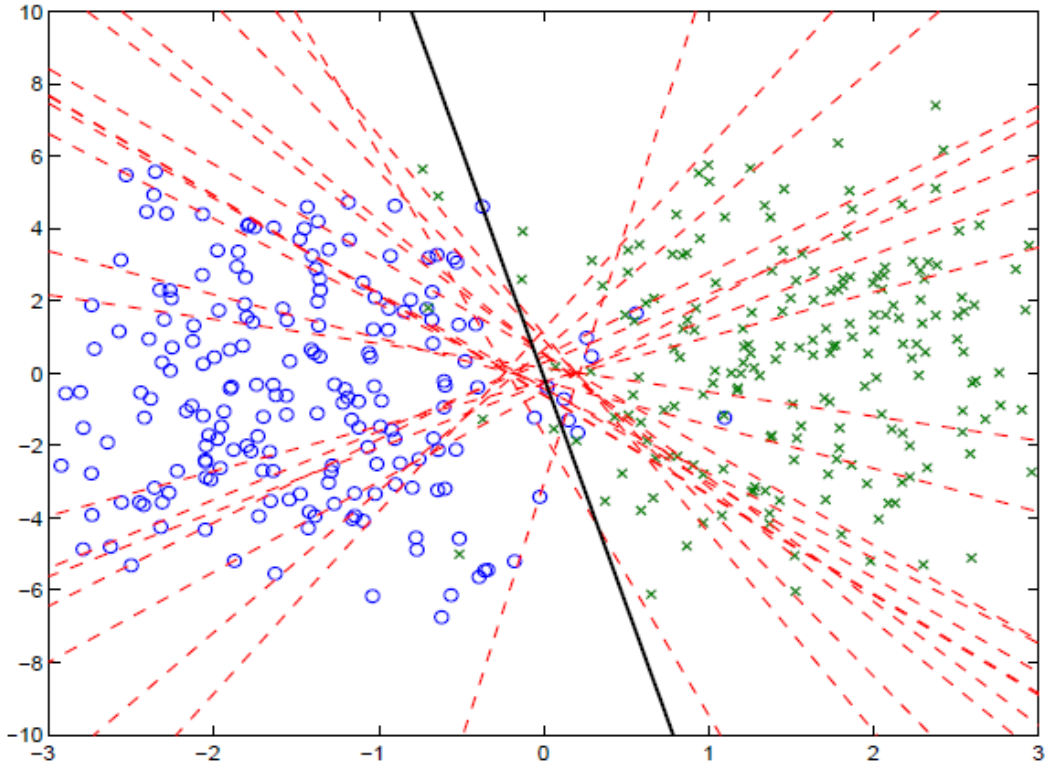
- Data $(\mathbf{a}_i, b_i), i = 1, \cdots, N$, $\mathbf{a}_i \in \mathbb{R}^K$, $y_i \in \{-1, +1\}$
- Linear SVM

$$\min_{\mathbf{w}} \frac{1}{N} \sum_{i=1}^{N} \max\left\{0, 1 - b_i \boldsymbol{x}^T \mathbf{a}_i\right\} + \frac{\lambda}{2} \|\boldsymbol{x}\|_2^2 \tag{0.5}$$

- Example: $K = 2$, $N = 400$, splits into $20$ groups

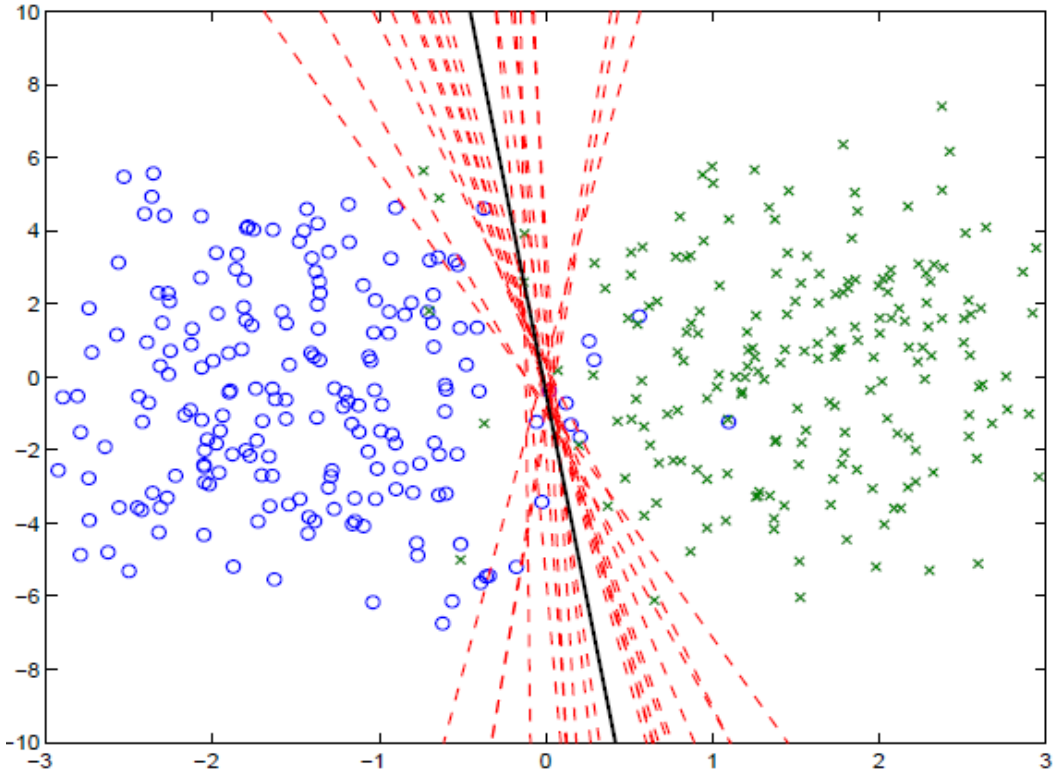
# Consensus SVM (from Boyd)

## Iteration 1



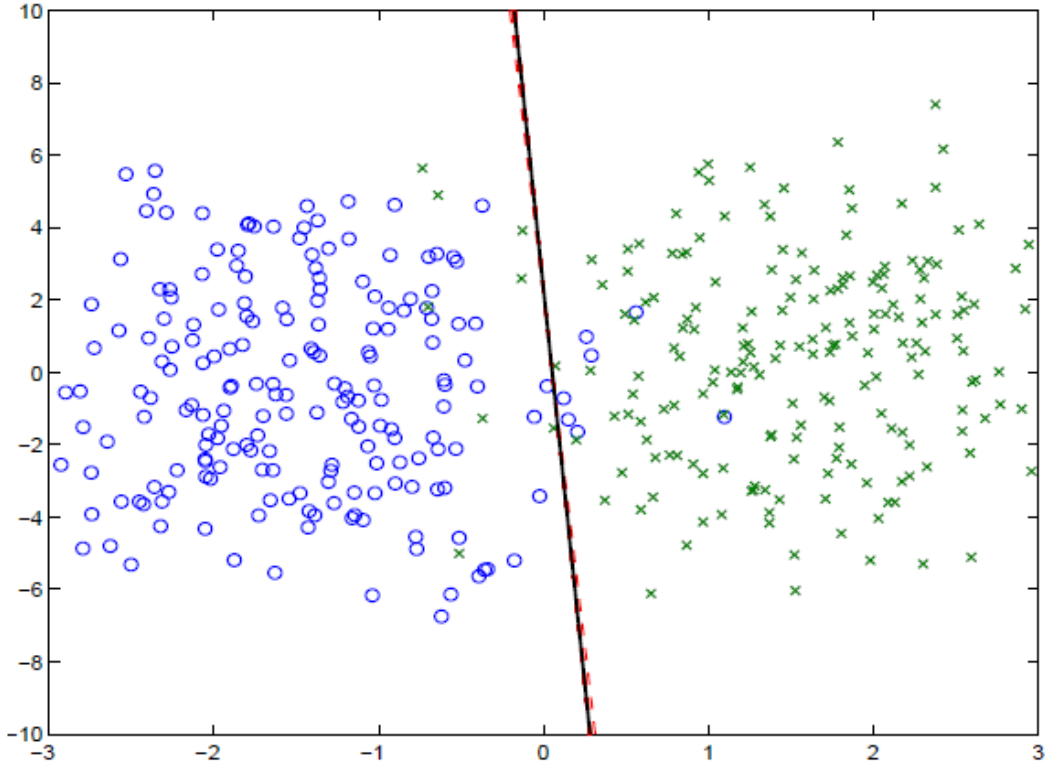
# Consensus SVM (from Boyd)

## Iteration 5



# Consensus SVM (from Boyd)

## Iteration 40



# Application 5: Sharing Problem

- Consider the following sharing problem

$$\min \quad \sum_{i=1}^{N} f_i(\boldsymbol{x}_i) + g\left(\sum_{i=1}^{N} \boldsymbol{x}_i\right) \tag{0.6}$$

- $f_i$ local cost function; $g$ shared objective
- Important class of problem, applications for example in economics
- Agents have selfish interests, but their behaviors are coupled together by some sort of "mutual interactions"

# Application 5: Sharing Problem

- **First Approach** Introducing a new variable $\mathbf{z}$

$$\min \quad \sum_{i=1}^{N} f_i(\boldsymbol{x}_i) + g\left(\mathbf{z}\right)$$

$$\mathbf{z} = \sum_{i=1}^{N} x_i$$

- **Second Approach** Introducing a set of new variable $\{\mathbf{z}_i\}$

$$\min \quad \sum_{i=1}^{N} f_i(\boldsymbol{x}_i) + g\left(\sum_{i=1}^{N} \mathbf{z}_i\right)$$

$$\mathbf{z}_i = \boldsymbol{x}_i$$

# Outline

# Review: Dual Problem for Constrained Optimization

- Primal problem (with equality constraints)

$$(P): \min \quad f(\boldsymbol{x}), \quad \text{subject to } \mathbf{A}\boldsymbol{x} = \mathbf{b},\ \boldsymbol{x} \in X \qquad (0.1)$$

  $f(\cdot)$ is a convex function
- Consider the Lagrangian, $\mathbf{y}$ is the dual variable (multiplier)

$$L(\boldsymbol{x}, \mathbf{y}) = f(\boldsymbol{x}) + \langle \mathbf{y}, \mathbf{A}\boldsymbol{x} - \mathbf{b} \rangle$$

- Dual function

$$d(\mathbf{y}) = \min_{\boldsymbol{x} \in X} L(\boldsymbol{x}, \mathbf{y}) \leq \min_{\mathbf{A}\boldsymbol{x} = \mathbf{b}, \boldsymbol{x} \in X} f(\boldsymbol{x})$$

- Dual problem (unconstrained)

$$(D): \max_{\mathbf{y}} d(\mathbf{y}) = \max_{\mathbf{y}} \min_{\boldsymbol{x} \in X}\ L(\boldsymbol{x}, \mathbf{y}) \leq \min_{\mathbf{A}\boldsymbol{x} = \mathbf{b}, \boldsymbol{x} \in X} f(\boldsymbol{x})$$

- $f$ convex, $X$ convex set, strong duality, $P = D$ (optimal objective equals)

# Dual Ascent

- Dual objective is concave

$$(D): \max_{\mathbf{y}} d(\mathbf{y}) = \max_{\mathbf{y}} \min_{\boldsymbol{x}} L(\boldsymbol{x}, \mathbf{y}) = \max_{\mathbf{y}} \min_{\boldsymbol{x}} f(\boldsymbol{x}) + \langle \mathbf{y}, \mathbf{A}\boldsymbol{x} - \mathbf{b} \rangle$$

- Gradient Ascent for the dual problem

$$\mathbf{y}^{r+1} = \mathbf{y}^r + \alpha \nabla d(\mathbf{y}^r)$$

- Recall our dual SVM problem and algorithm?

# Dual Ascent (cont.)

- Let $\widetilde{\boldsymbol{x}} = \arg\min_{\boldsymbol{x}\in X} L(\boldsymbol{x}, \mathbf{y}^r)$, we know (can be verified)

$$\nabla d(\mathbf{y}^r) = \mathbf{A}\widetilde{\boldsymbol{x}} - \mathbf{b}$$

- **Dual Ascent Algorithm** (intuition?)

$$\boldsymbol{x}^{r+1} = \arg\min_{\boldsymbol{x}\in X} L(\boldsymbol{x}, \mathbf{y}^r)$$
$$\mathbf{y}^{r+1} = \mathbf{y}^r + \alpha\left(\mathbf{A}\boldsymbol{x}^{r+1} - \mathbf{b}\right)$$

# Dual Decomposition

- If $f$ is separable among variables

$$(P): \min_{\sum_{i=1}^N \mathbf{A}_i \boldsymbol{x}_i = \mathbf{b}} f(\boldsymbol{x}) = \sum_{i=1}^N f_i(\boldsymbol{x}_i) \tag{0.2}$$

- Consider the Lagrangian

$$L(\boldsymbol{x}, \mathbf{y}) = \sum_{i=1}^N L_i(\boldsymbol{x}_i, \mathbf{y}) = \sum_{i=1}^N f_i(\boldsymbol{x}_i) + \langle \mathbf{y}, \mathbf{A}_i \boldsymbol{x}_i - \mathbf{b} \rangle \tag{0.3}$$

- Dual ascent algorithm (a.k.a. dual decomposition)

$$\boldsymbol{x}_i^{r+1} = \arg\min_{\boldsymbol{x}_i \in X_i} L_i(\boldsymbol{x}_i, \mathbf{y}^r), \ \forall \ i$$

$$\mathbf{y}^{r+1} = \mathbf{y}^r + \alpha \left( \sum_{i=1}^N \mathbf{A}_i \boldsymbol{x}_i^{r+1} - \mathbf{b} \right)$$

- Large-scale problem: Split variables, update $\boldsymbol{x}_i$ in parallel

# Convergence of Dual Decomposition

- Assumption for convergence
  1. Dual function $d(\mathbf{y})$ has Lipschitz continuous gradient (with constant $L$)
  2. The stepsize $\alpha \leq \frac{1}{L}$
- (1) requires that problem (P) is strongly convex with constant at least $\frac{1}{L}$
- Consider again the dual ascent algorithm we tried for the SVM problem

# Method of Multipliers

- Primal problem (with equality constraints)

$$(P): \min \quad f(\boldsymbol{x}), \quad \text{subject to } \mathbf{A}\boldsymbol{x} = \mathbf{b},\ \boldsymbol{x} \in X \tag{0.4}$$

  $f(\cdot)$ is a convex function
- Consider the Augmented Lagrangian, $\mathbf{y}$ is the dual variable (multiplier)

$$L(\boldsymbol{x}, \mathbf{y}) = f(\boldsymbol{x}) + \langle \mathbf{y}, \mathbf{A}\boldsymbol{x} - \mathbf{b} \rangle + \frac{\rho}{2} \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|^2$$

- $\rho > 0$ is penalty parameter
- Quadratic penalty term penalizes constraint violation
- **Method of Multipliers** (intuition?)

$$\boldsymbol{x}^{r+1} = \arg\min_{\boldsymbol{x} \in X} L_\rho(\boldsymbol{x}, \mathbf{y}^r)$$

$$\mathbf{y}^{r+1} = \mathbf{y}^r + \rho \left( \mathbf{A}\boldsymbol{x}^{r+1} - \mathbf{b} \right)$$

# Method of Multipliers

- **Good**
  1. Weak Assumptions: $f$ only needs to be convex
  2. Dual function has Lipschitz continuous gradient
- **Bad**
  1. Objective not separable anymore
  2. Dual decomposition no longer works
- ADMM combines
  1. Method of multiplies: Use augmented Lagrangian
  2. Dual decomposition: Can do variable splitting